

GENERAL BIOLOGY

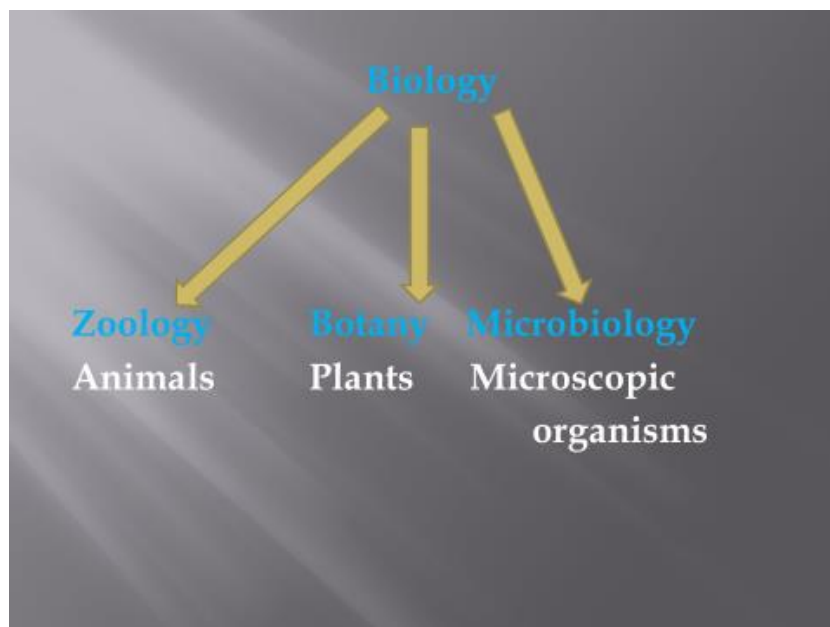
Introduction

Nature and Scope of Biology

- ▣ Biology is the study of living things
- ▣ It is the Science of life
- ▣ It has formed by 2 Greek words
- ▣ Bios means – life
- ▣ Logos means – knowledge

Biology and its branches

- ▣ Traditionally two branches –Botany and Zoology
- ▣ 3rd division – Microbiology
- ▣ Study of microscopic animals not in Zoology and Botany



Subjects studied in various branches of Biology

- ▣ Physiology – Organ systems
- ▣ Embryology – Formation of gametes, Events from fertilization of egg to the development of embryo
- ▣ Ecology – Interaction of plants and animals with the environment.
- ▣ Taxonomy – Identification, nomenclature and classification of organisms.
- ▣ Evolution – Continuous adaptations of plants and animals to the environment.
- ▣ Genetics – Hereditary and variations

Course Capsule

- ▣ The Nature of Biology:
- ▣ The branches of Biology, Methods of inquiry in Biology, some basic concepts, Applied Biology.
- ▣ Biomes to Microhabitats.
- ▣ Communities and Ecosystems.
- ▣ Pollution and Conservation.
- ▣ Systematics: The taxonomic hierarchy, Naming of organisms.
- ▣ Classification of Organisms: Prokaryota, Monera, Protista, Fungi, Plantae, Animalia.
- ▣ The Cell as a basic unit: Fine structure of Prokaryotic and Eukaryotic cells.
- ▣ Tissues, Organs and Individuals: Animal tissues, Plant tissues, Organs and organ systems.
- ▣ Nutrition: Autotrophic and heterotrophic nutrition, Photosynthesis.
- ▣ Respiration: Aerobic and anaerobic respiration, BMR.
- ▣ Transport in animals and plants.
- ▣ Defense against diseases: Defense systems in animals, Plant defenses against diseases.
- ▣ Reproduction: Asexual and sexual reproduction.
- ▣ Principles of Heredity: Monohybrid inheritance, Dihybrid inheritance.
- ▣ Evolutionary trends.

WEEK 01

Biology and its relationship with other sciences.

- ▣ To study Biology we should know other sciences like **physics** and **chemistry**.

CHEMISTRY

- Cell membrane – structure of lipids and proteins
- Release of energy during respiration –Electron transport chain

PHYSICS

- Effect of light on flowering – Absorption of different wave lengths of light

Scientific method in Biology

- ▣ It is difficult to find an answer without logical solution.
- ▣ We need to define the problem, experiment it, analyze it and come to a conclusion
- ▣ It is done through Scientific method which draws a definite conclusion



1. Observation

- **Most biological investigations:** – start with observation on Structure, process, behavioral pattern etc.



2. Defining a problem

- ▣ The observations lead to questions:
- ▣ What is this, how does it work, why does it work the way it does?
- ▣ This may necessitate further observations to be made

3. Hypothesis

- **On the basis of the facts known already** – make a tentative theory
 - It is an intelligent guess
 - It is a gradual accumulation of indirect evidence
 - It may have number off predictions

4. Controlled experiment

Test each prediction

- ▣ It is geared specifically towards rejecting one's favored hypothesis
- ▣ Often, two experiments are conducted at the same time.
- ▣ In one experiment, all the variables are kept constant except one
- ▣ The other experiment - **control experiment** -variable is left unaltered.
- ▣ The results of the two experiments are compared to determine if the tested variable has an effect on the outcome.

5. Theory

- ▣ When Hypothesis is supported by evidence it becomes a theory
- ▣ More evidence, more acceptance confirms it
- ▣ It is not a fact that cannot be changed

- ▣ Ex. Theory of evolution – changed over time

Role of Biology

Objectives of studying Biology

1. Understand ourselves better

- We study basic concepts related to structure and functions of cells and organs

2. Give an idea about our place in the universe

- Man is a small part of the universe We need to protect and respect other living things

3. Meet our needs

- Helps to utilize resources available.
- Ex. Medicine- cure diseases

4. **Warn about health hazards** due to use of pesticides, fertilizers

5. **Increase awareness** of interrelationships of organisms with environment.

Properties of life

- ▣ Biology is the study of living things
- ▣ What makes something alive?
- ▣ All living organisms share 5 basic characteristics

1).Order

- All organisms are composed of one or more cells in a highly ordered structure
- Atoms-molecules-cellular organelles- cells

2). Sensitivity

- Living things respond to stimuli
- Ex. Plants grow towards light

3). Growth, Reproduction, development

- Living things are capable of growing and reproducing
- They Possess hereditary molecules and are passed to offspring

4). Homeostasis

- Capable of keeping internal environment constant

5). Evolution

- With the time living things change according to the environment

Week 02

The Cell – As the basic unit of life

Cell is the basic unit of structure and function of living things



CELL THEORY

- ▣ Proposed by Schlieden and Schwan in 1838
- ▣ It includes 3 principles
 - All organisms are composed of one or more cells and metabolism and hereditary occur within these cells.
 - Cells are the smallest living things, the basic units of organization.
 - Cells arise only by division of previously existing cell.

Cell as a self-contained unit

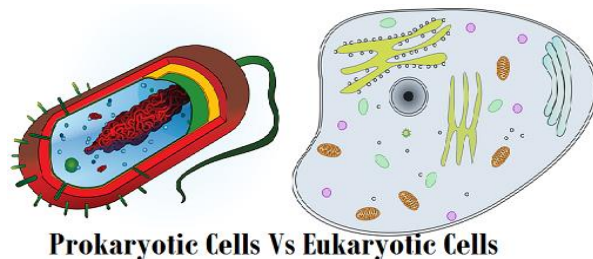
- ▣ Cell though very small, it acts as an autonomous unit.

Cells;

- 1). Carry out all vital biological processes.
- 2). oxidize the food molecules - produce energy and store these energy rich molecules
- 3). Build new structures and can replace worn out cells by using nutrient molecules.
- 4). repair and exchange gases with its surrounding
- 5). replicate its own self
- 6). have their own lifespan

Prokaryotic and Eukaryotic cell

- According to the cell structure there are two types of cells
- i. **Prokaryotic cell**
- ii. **Eukaryotic cell**

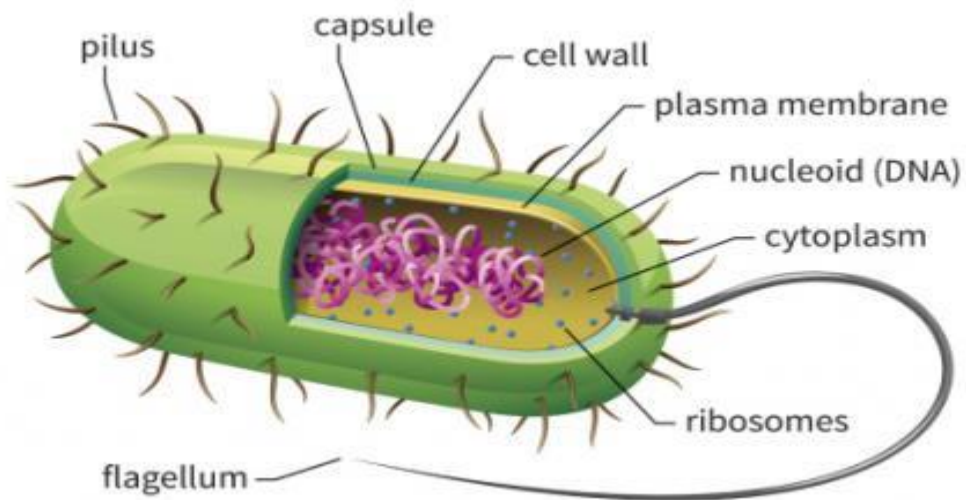


1).Prokaryotic Cell

- ▣ Prokaryotes is the term given to organisms without true nucleus.
- ▣ Hereditary material called DNA is not enclosed in a membrane / no membrane bounded internal structures.

Structure of prokaryotic cell

- ▣ We consider Bacterial cell as an example for Prokaryotic cell Bacteria cell is composed of different parts.



1).Cell wall

- ▣ Most bacteria are encased by a strong cell wall made up of peptidoglycan.
- ▣ No Eukaryotes possess cell walls with this type of chemical composition.

2).Capsule

- ▣ The cell wall is covered by a capsule or slim layer.
- ▣ It gives additional protection to the cell from unfavorable conditions.
- ▣ Helps to stick to each other and form colonies.

- ▣ Helps to stick to the surface of the food supply or to the host cell.

3).Plasma membrane

- ▣ Below the cell wall there is a membrane called plasma membrane.
- ▣ It is selectively permeable. It permits some molecules to pass through it. But not others.

4).Cytoplasm

- ▣ Plasma membrane encloses the cytoplasm.
- ▣ It looks granular due to the presence of ribosome.
- ▣ There are enzymes, carbohydrates, lipids and genetic materials can be found in the cytoplasm.

5).Flagella

- ▣ Many bacteria are equipped with one or more flagella.
- ▣ It helps bacteria to move and to respond to stimuli.

2).Eukaryotic cell

- They are highly evolved cells possessing specialized organelles, a definite membrane bounded nucleus and highly organized cytoplasm.
- They show great variations in shapes, size and type.
- It is composed of jelly like translucent protoplasm bound by a plasma membrane.

▣ Protoplasm has 2 parts,

- Cytoplasm
- Nucleus

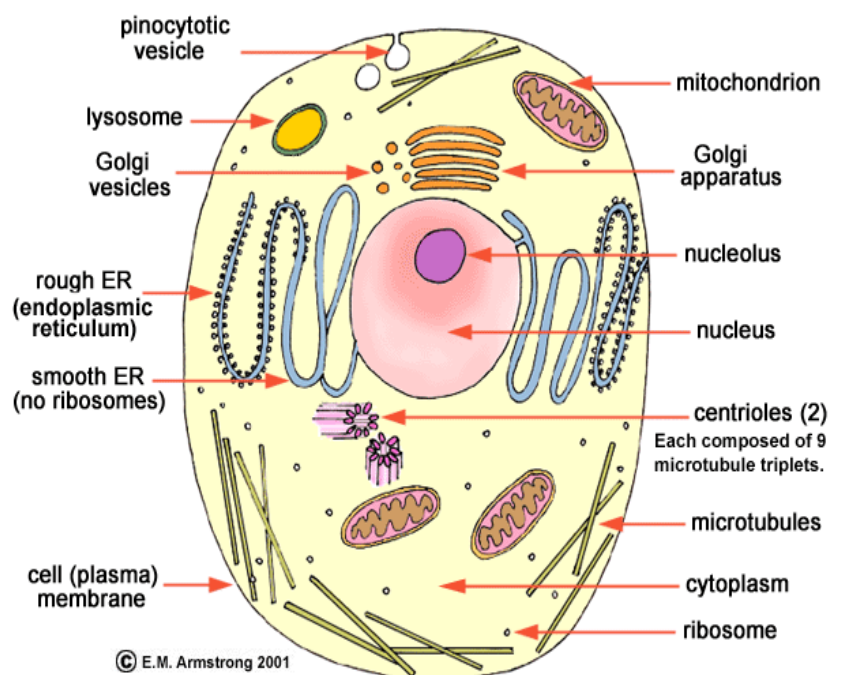
▣ Cytoplasm has number of compartments.

They are called as organelles and carryout different functions.

▣ These organelles may or may not be bound by a membrane

Ex:

Mitochondria
Ribosome
Golgi
Apparatus
Lysosome

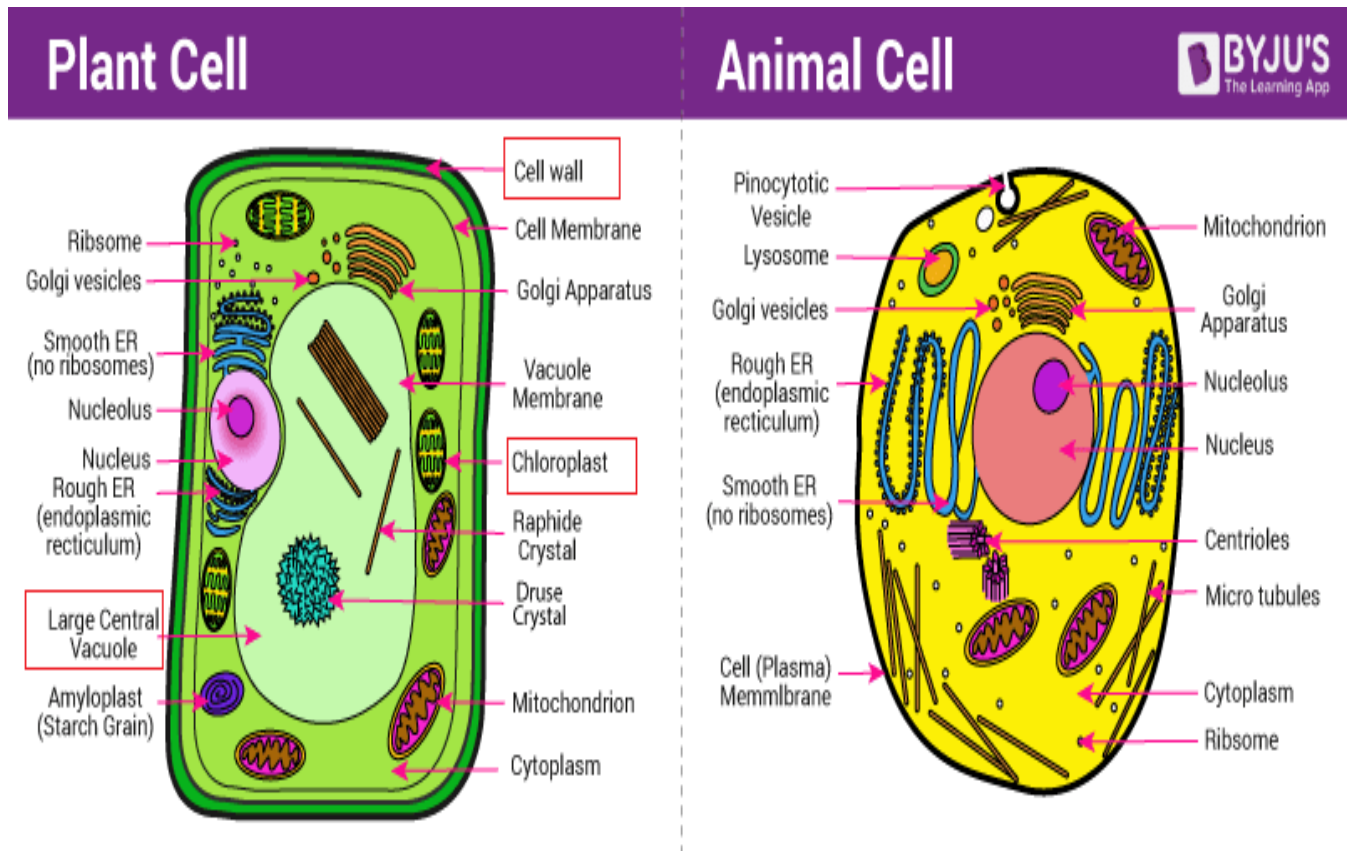


Types of Eukaryotic cells

1. Plant cell
2. Animal cell

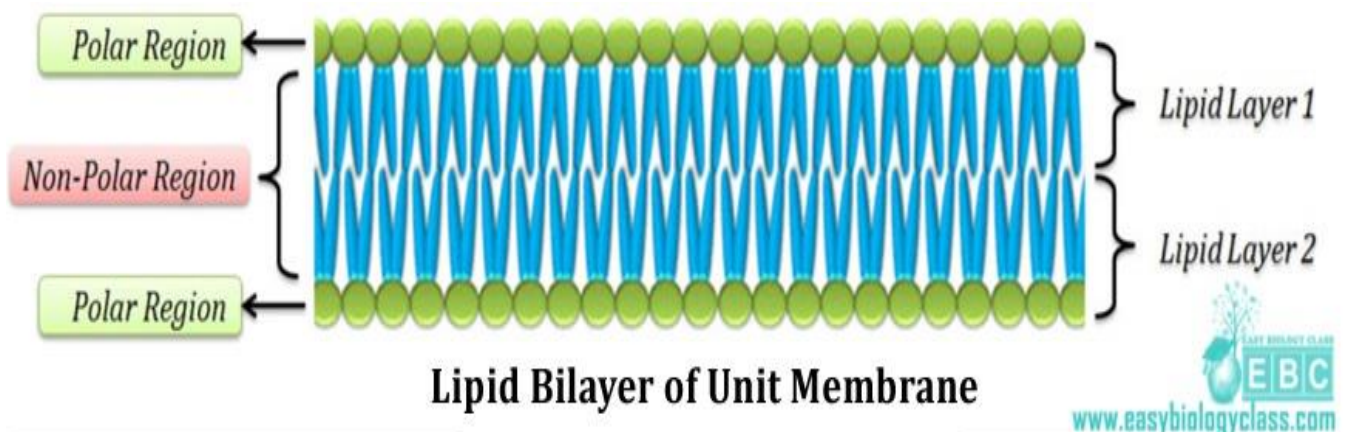
Major differences between plant cell and animal cell are;

1. Plant cell contains a cell wall but animal cell does not
2. Plant cell contains a large central vacuole but animal cell contains number of small vacuoles
3. Plant cell contains a chloroplast but animal cell does not

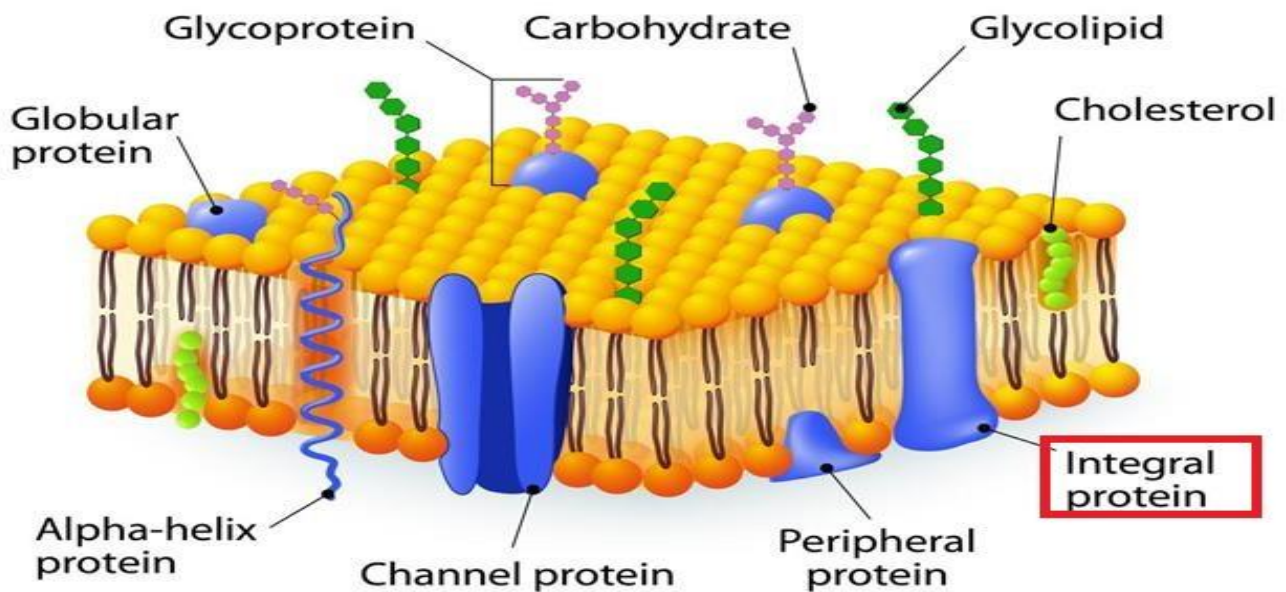


Structure of the Plasma membrane

- ▣ The most favored model is the fluid mosaic model.
- ▣ Proposed by Singer and Nicholson in 1972
- ▣ Plasma membrane is made up of lipid, protein and carbohydrates
- ▣ Lipids form a fluid bilayer
- ▣ In this bilayer, heads of lipid molecules are directed outwards and they are polar and hydrophilic
- ▣ Tails are directed inwards and they are non-polar and hydrophobic.



Proteins are two types



1). Intrinsic proteins/ Integral Protein

Embedded in the lipid bilayer.

- ▣ They are held by hydrophobic and hydrophilic parts of the lipid molecules and cannot be easily dislodged.
- ▣ Some protein molecules are small and only a part projects at the surface.
- ▣ Some are large and projects at both surfaces.

2). Extrinsic proteins / peripheral protein

Are superficially attached to the lipid layer. They can be easily removed.

Carbohydrates forms the antennae projecting on the surface.

The cells recognize one another by these.

1).Cell wall

i. Primary cell wall

▣ Is laid on the middle lamella within the cell.

▣ It consists of cellulose micro fibrils running through the matrix.

ii. Secondary cell wall

▣ It is built up by addition of cellulose layers on the inside surface of primary cell wall.

iii. Plasmodesmata

▣ The cell wall does not form a continuous layer and is interrupted by minute pores called plasmodesmata.

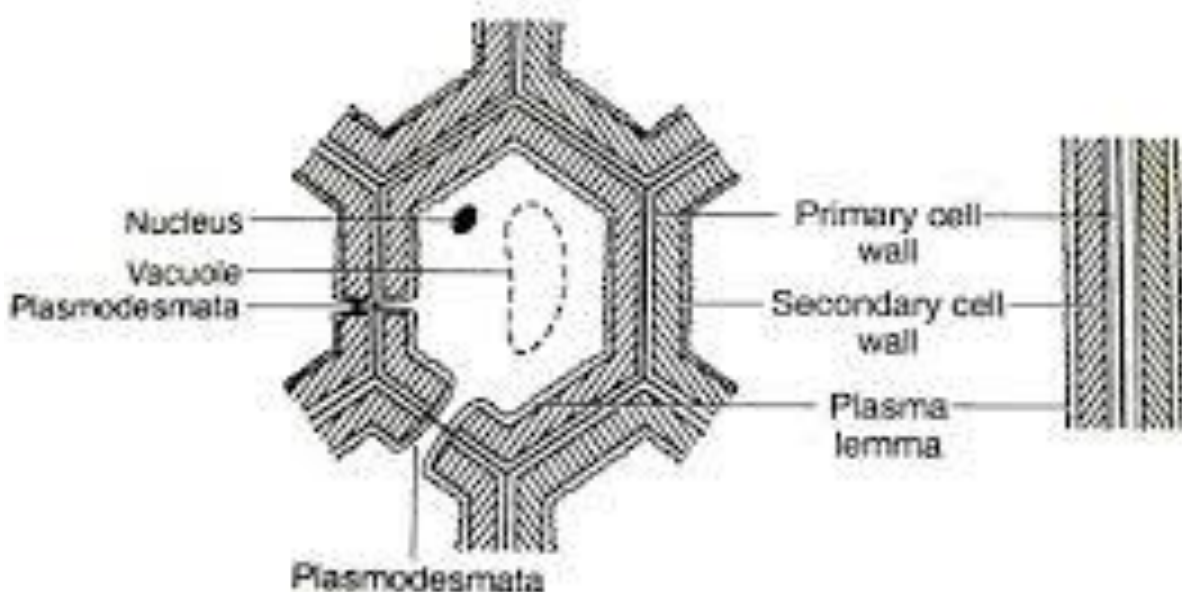


Fig - Arrangement of various layers of cell wall in a mature plant cell.

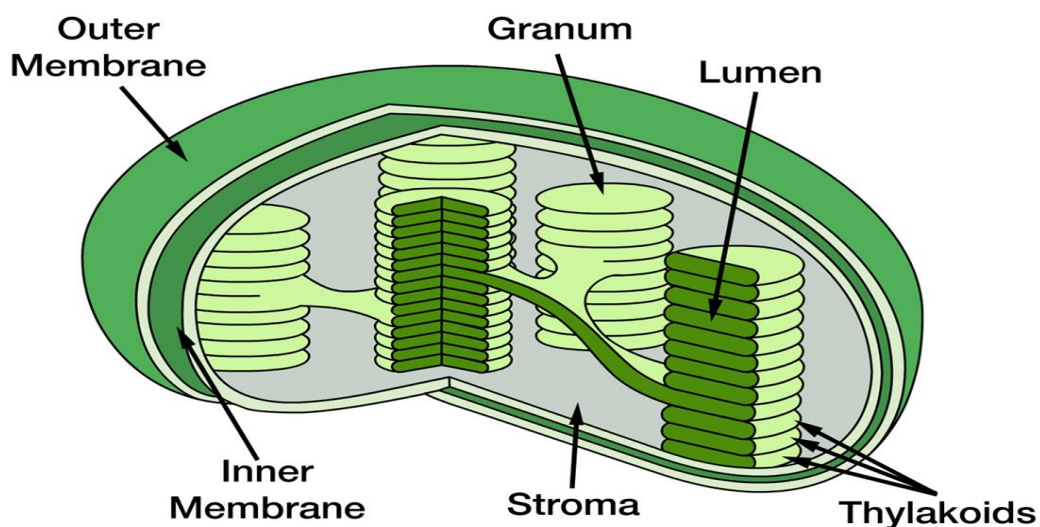
Week 05

Cellular Organelles

Chloroplast

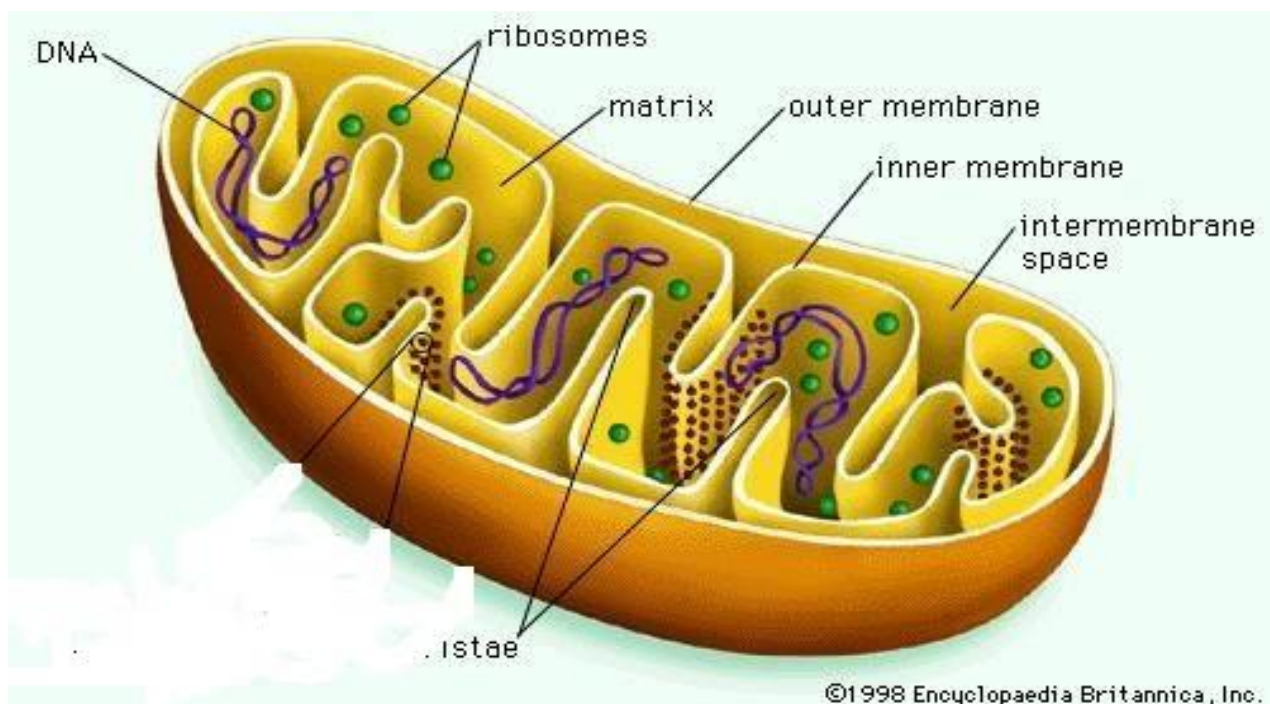
- ▣ It is surrounded by 2 membranes.
- ▣ These 2 membranes are separated by a space.
- ▣ These membranes enclosed a fluid space – called stroma / matrix.
- ▣ A membrane system is suspended in the matrix.
- ▣ It carries out many flattened, fluid filled sacs called thylakoids or lamella.
- ▣ These thylakoids stack to form grana.
- ▣ Various grana are jointed by integrated lamella.

Chloroplast



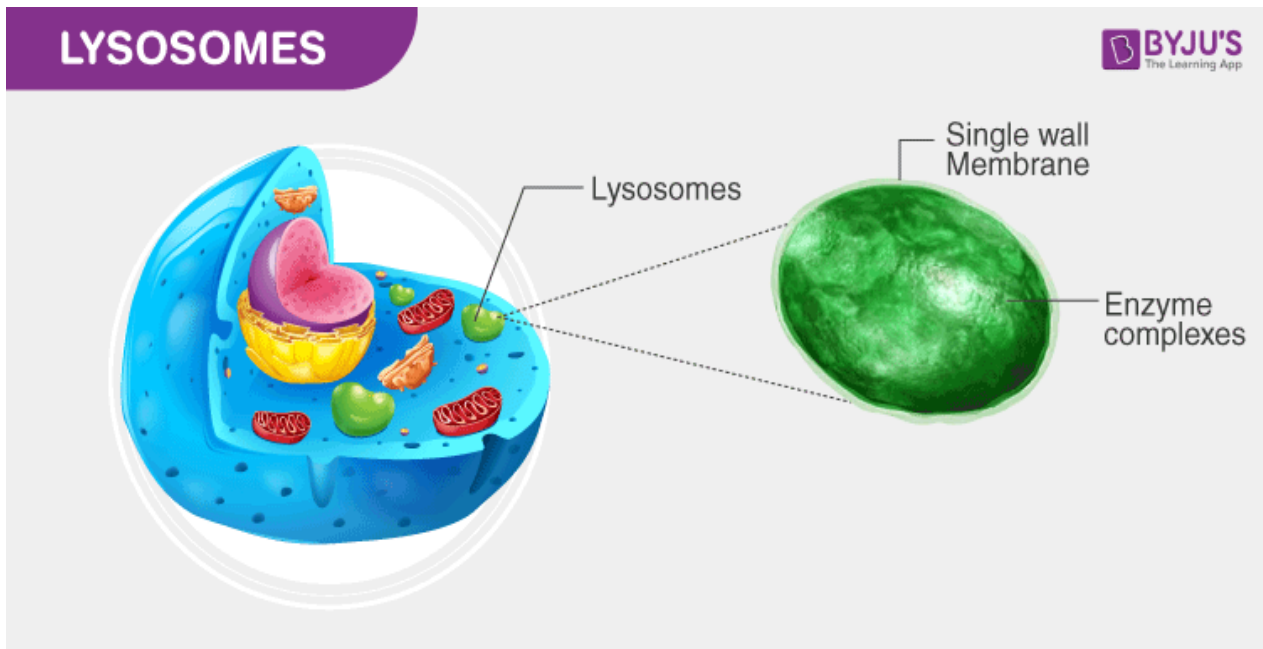
Mitochondria

- ▣ This is called as power house of the cell.
- ▣ It is specialize in the synthesise of ATP, the usable chemical energy.
- ▣ It is sausage shaped.
- ▣ It is bound by 2 membranes that separated by a space.
- ▣ Inner membrane folds into numerous folds Called cristae.
- ▣ Inner surface of the cristae is packed with elementary particles / granules.
- ▣ Matrix is lying inside the inner membrane.



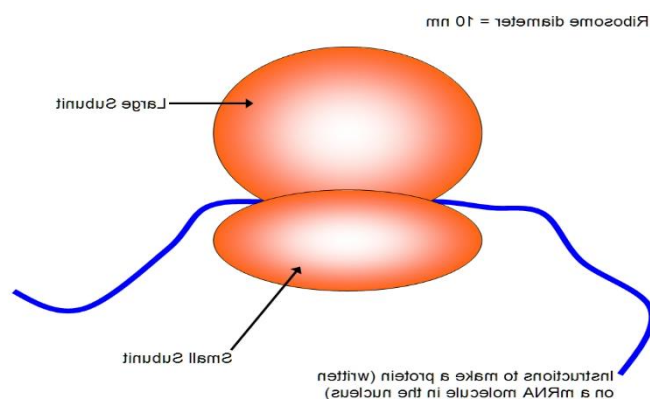
Lysosome

- ▣ They are membrane bound digestive vesicles.
- ▣ Lysosomes are bound by a single membrane.
- ▣ They contain a mixture of digestive enzymes



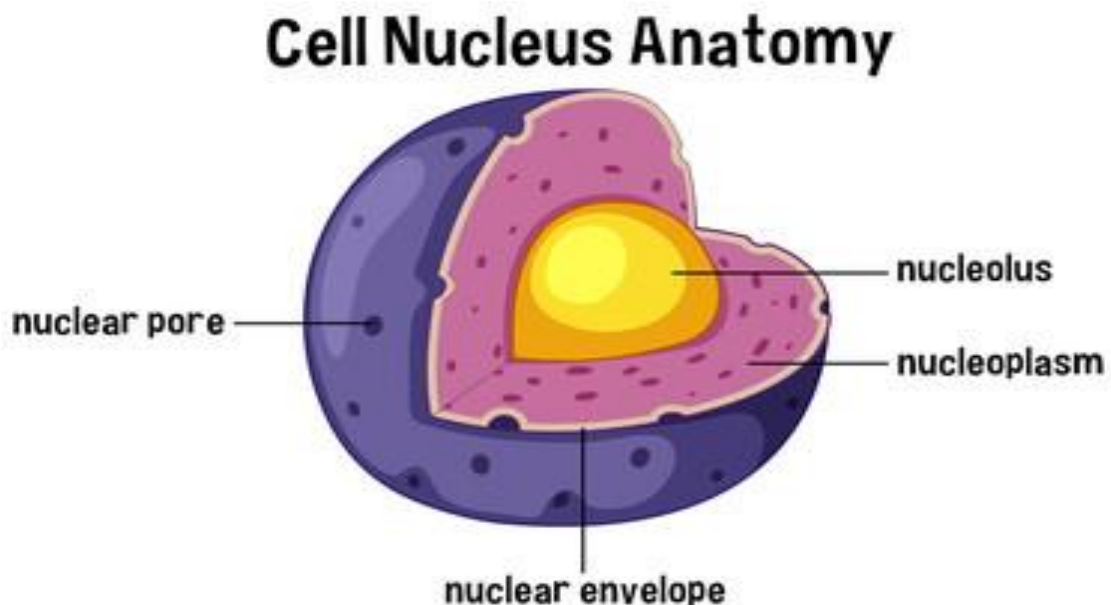
Ribosome

- ▣ They are spherical bodies found in the cytoplasm of the cell.
- ▣ It is composed of large and small subunit.
- ▣ They serve as sites of protein synthesis.



Nucleus

- ▣ This is the largest and most easily seen organelle
- ▣ This is the core of the cell which command all metabolic activities.
- ▣ In animal cells, nucleus is located in the central region.
- ▣ Large amount of DNA is packed in the nucleus.
- ▣ Roughly spherical in shape.
- ▣ The nucleus is Encased by 2 membranes
- ▣ Nucleolus is the site within the nucleus where ribosome are made.
- ▣ It contain large amount of DNA and RNA
- ▣ The space between the nuclear envelops and the nucleolus is filled by the transparent, granular nucleoplasm.



Week 06

Classification of life

- ▣ In early history people knew only **plants and animals**
- ▣ Linnaeus developed his system of classification, there were only two kingdoms, **Plants and Animals**.
- ▣ The invention of the microscope led to the discovery of new organisms and the identification of differences in cells.
- ▣ So, the two-kingdom system was **no** longer useful.
- ▣ Today the system of classification includes **six kingdoms**.

The six kingdoms of life

1. Archaeobacteria
2. Eubacteria (bacteria and cyanobacteria / blue green algae)
3. Protista (algae and slime moulds)
4. Fungi (mushroom, yeasts, moulds)
5. Plantae (plants)
6. Animalia (animals)

- ▣ In addition to this, there are '**viruses**'.
- ▣ These are sub-microscopic agents that can be **seen only through the electron microscope**.
- ▣ They are **unable to grow and reproduce outside living cells**.
- ▣ They share both living and non-living characteristics.

Living and non living characteristics of Viruses

Living

- Possess genetic material DNA/RNA
- Possess protein coat
- Can replicate (inside a living cell)

Non-Living

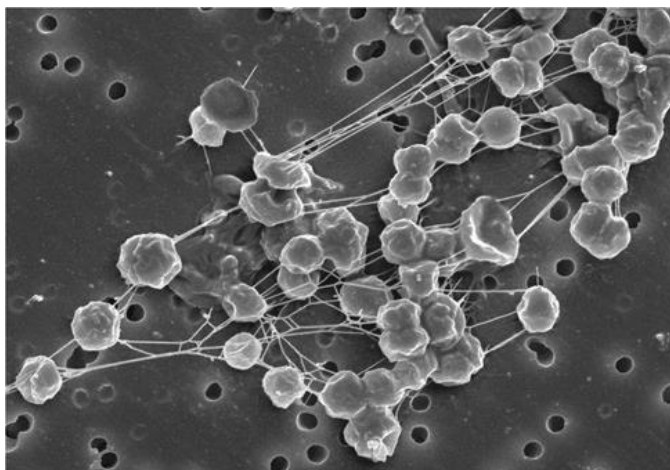
- Non cellular
- Cannot reproduce by themselves
- Do not possess ribosomes, mitochondria etc
- Only have one type of nucleic acid

Kingdom 1. Archaeobacteria

- ▣ These are a diverse group of bacteria
- ▣ Archaeobacteria are different from Eubacteria, from the places where they live.
- ▣ They live in **extreme conditions which are not suitable for other normal life forms**
- ▣ There are **three major known groups of Archaeobacteria**:
 - i. Methanogens
 - ii. Halophiles
 - iii. Thermophiles

1). Methanogens

- ▣ They live in **oxygen less environments**
- ▣ They are found in sewage treatment plants, bogs, and the intestinal tracts of some animals.



2). Halophiles

- ▣ Are bacteria that thrive in high salt concentrations such as sea water



3). Thermophiles

- ▣ Are the heat-loving bacteria found in hot water springs?

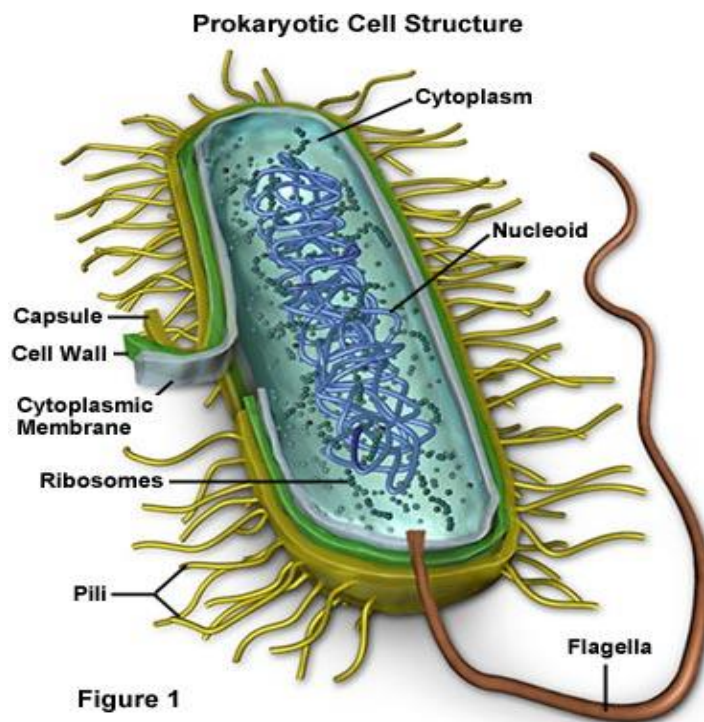


Kingdom 2. Eubacteria

- ▣ Includes Bacteria and Cyanobacteria

Bacteria

- ▣ Tiny minute organisms that cannot be seen with the naked eye



Cyanobacteria

- ▣ These are prokaryotic, photosynthetic (can make their own food) simple life forms

Ex. Microcystis

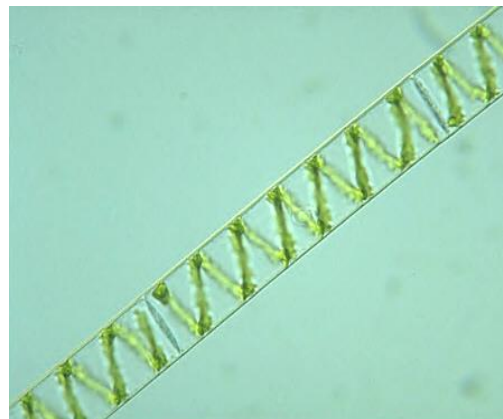
Week 07

Kingdom 3. PROTISTA (Algae & slim moulds)

- ▣ All are Eukaryotes
- ▣ All algae contain **Chlorophyll a**
- ▣ Depending on the type of algae along with chlorophyll 'a' they can have '**chlorophyll b**', '**c**', '**d**' or '**e**'
- ▣ Thus these algae are all **photosynthetic protista**.
- ▣ Algae may occur in oceans, lakes, tanks, rivers streams and also in damp soil and on bodies of other organisms.

Ex. *Clamidomonas*

Spirogyra



Kingdom 4. Fungi

- ▣ **non-photosynthetic** (no chlorophyll so can't produce their own food), eukaryotic organisms
- ▣ **heterotrophic** (get their food from others)
- ▣ Most are small, microscopic, filamentous branched forms that bear spores

Ex. Mushrooms

Kingdom 5. Plantae

- ▣ Two types of plants as vascular or non-vascular
- ▣ Non-vascular plants are more primitive

Ex.

Thalloid liverworts

Mosses

ferns

Vascular plants Can be of two types,

1. Open seeded plants (Ex Cycas)
2. Enclosed seeded plants

Can be

- Dicot plants
- Monocot plants

Differences between dicot and monocot plants

Monocots				
				
One cotyledon	Veins usually parallel	Vascular bundles usually complexly arranged	Fibrous root system	Floral parts usually in multiples of three
Embryos	Leaf venation	Stems	Roots	Flowers
Dicots				
				
Two cotyledons	Veins usually netlike	Vascular bundles usually arranged in ring	Taproot usually present	Floral parts usually in multiples of four or five

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Kingdom 6. Animalia

This kingdom consists of

- ▣ multicellular (composed of more than one cell)
- ▣ heterotrophic (feed on other organisms – can't produce their own food)organisms

Divided into 2 main Groups

1. Invertebrates
2. Vertebrates

Invertebrates

Consists of,

- Porifera –Sponges
- Cnidaria
- Plathihelminthes
- Nematoda
- Annelida
- Mollusca
- Arthropoda
- Echinodermata

Vertebrates

Consists of

- ▣ Pisces - Fish
- ▣ Amphibia
- ▣ Reptilia
- ▣ Aves - Birds
- ▣ Mammalia

Week 08

Phylum: Porifera

- ▣ Called pore bearers as their body wall contain minute pores.
- ▣ All are aquatic marine or fresh water.

Ex. *Euspongia*



Phylum Cnidaria

- ▣ Radial symmetrical animals
- ▣ They have specialized cells called Cnidocytes – “stinging cells”

Ex. jellyfish

Phylum Platyhelminthes (Flatworms)

- ▣ Soft, dorso-ventrally flattened animals with bilateral symmetry
- ▣ Most are parasitic.

Ex. *Taenia solium*



Phylum Nematoda

- ▣ Commonly known as round worms
- ▣ Occur almost everywhere in soil, fresh water, salt water and as parasites in the body
- ▣ Body wall is covered by a tough cuticle

Ex: *Ancylostoma*- Hook worm infection



Phylum Annelida

- ▣ Elongated segmented body.
- ▣ Body is covered by a transparent cuticle.

Ex. *Hirudinaria* – Leech



Phylum Mollusca

- ▣ Live in fresh water, marine water and on land.
- ▣ Soft bodied.
- ▣ Have a muscular foot used for locomotion

Ex. Snail



Phylum Arthropoda

- Largest group of animals

Reasons for the success of Arthropods,

1).Chitinous exoskeleton

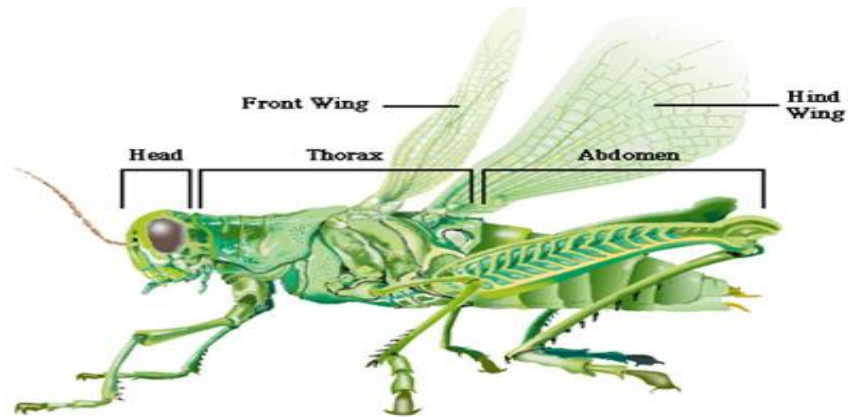
- ▣ They have a unique chitinous (made up of chitine) cuticle which is light weight and tough.
- ▣ It protects the animals from predators.
- ▣ It is impervious to water and so protects the body from drying and has made the arthropods survive on land in almost all types of environments.
- ▣ Insects have chitinous wings that help them to fly.
- ▣ The biting mouth parts, lenses for the eye, and sensory organs are also made up of chitin.

Chitinous exoskeleton of beetle



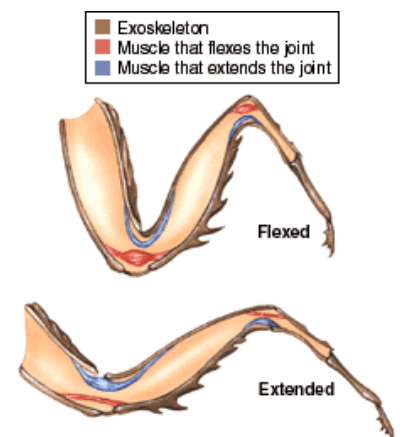
2. Segmentation

- ▣ Body is segmented into distinct regions as head, thorax, and abdomen; each one is specialized to perform a specific function.



3).Jointed appendages

- ▣ The appendages have been modified to perform different functions.
- ▣ It increases the efficiency and complexity of the activity.
- ▣ It is also help the arthropods to adjust to different habitats.



4).Ecdysis/molting

- ▣ The chitinous skeleton does not allow the body to grow and hence to overcome it, ecdysis has evolved.
- ▣ The animal shed off the exoskeleton periodically and grows larger and secretes the new exoskeleton.

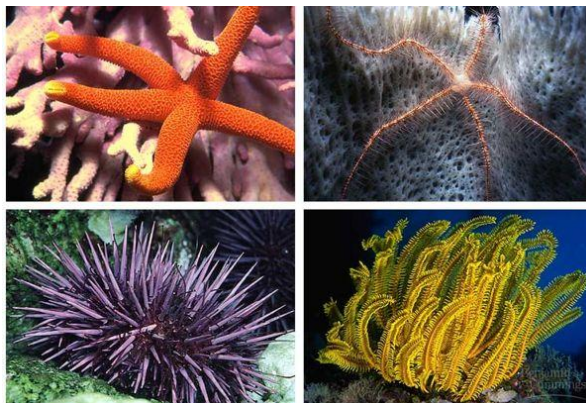
EX: Ecdysis of scorpion



Phylum Echinodermata

- ▣ All are marine animals
- ▣ Spiny skinned animals. The spines fixed or movable.
- ▣ Radial symmetrical animals with a water vascular system
- ▣ Have an anterior- posterior flattened body.

Ex; Echinoderms



Week 09

Phylum Chordata

Class – Pisces (Fishes)

Two types of fish,

- i. Cartilaginous fishes
- ii. Bony fishes

Cartilaginous fishes

- Cartilaginous fishes have cartilaginous skeleton
- They do not have a swim bladder so that they must keep swimming
- They are viviparous – live bearers.



○



Bony fishes

- have bony skeleton
- have spindle shaped body
- Respire through gills

- have scales in the body
- have a swim bladder help them to rise and sink
- have two chambered heart



Class - Amphibia

- ▣ They live in water and on land
- ▣ They have smooth and moist skin
- ▣ No scales present
- ▣ They have two pairs of limbs
- ▣ Their heart is three chambered.

Ex: Tree frog



Class – Reptilia

- ▣ They have dry horny scaly skin
- ▣ They have two pairs of limbs
- ▣ Their heart is three chambered

Ex: Python



Class – Aves

- ▣ Distinct feature of birds is flight

Characteristic features that help the birds in flight

- ▣ They have a stream lined body covered by feathers to help to reduce the resistance.
- ▣ The fore limbs are modified into wings. They are also covered by feathers
- ▣ Have light, porous bones with air cavities to reduce the weight
- ▣ Have a strong breast bone called keel to which the flight muscles are attached

- ▣ Heart is four chambered which allows complete separation of the oxygenated blood from the deoxygenated blood.
- ▣ Hind limbs are adapted for perching
- ▣ The tail is used for changing the direction
- ▣ Have a constant body temperature called warm blooded.
- ▣ They maintain a high metabolic rate in order to lift off the body from the ground
- ▣ The heart of the birds beats faster in order to supply large amount of oxygenated blood into the body
- ▣ Have large eyes and strong sense of sight. The field of vision is also great.
- ▣ They have a remarkable ability to judge the distance from a close range as well as from a great height
- ▣ The disappearance of right ovary and right oviduct reduces weight and is an adaptation for flight.



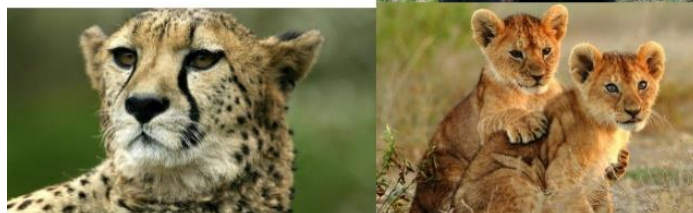
Class – Mammalia

- ▣ They are viviparous and nourish the young ones with milk secreted from mammary glands
- ▣ Body is covered with hair
- ▣ External ear present
- ▣ Spiny ant eater is an exceptional that Egg laying mammal

Remarkable mammals

- Spiny ant eater is an **egg laying mammal**
- In kangaroo, young ones are 3cm long when born. They crawl into the pouch and feed on the milk
- Shrews are the smallest, insectivorous mammals
- Bats are flying mammals. Their fore limbs are modified into wings
- Whales, Dolphins are aquatic mammals. Fore limbs are modified in to flippers
- Man, Monkey, Gorilla are highly evolved mammals. They have opposable thumb. Intelligent animals

Mammals



Ecology 01

Communities and Ecosystems

What is “Community” ?

All the organisms that live together in a place is called a community

What is “Co-evolution”

- ▣ The plants, animals, protists, fungi bacteria ect. live together and have changed and adjusted to one another continually over a period of millions of years.
- ▣ Such interactions are called coevolution

Symbiotic relationships

- Are shown by two or more kinds of organisms live in a place in permanent relationships.

Kinds of symbiosis

- commensalisms
- Mutualism
- Parasitism

Commensalism

- ▣ This relationship gives benefits for one species and neither hurts nor helps the other.
- ▣ Individuals of one species is physically attached to members of another.
- ▣ Ex. Epiphytes – They are the plants that grow on the branches of other plants.
- ▣ Host plant is unharmed and the epiphyte benefits by having the substrate for growing.

Mutualism

- ▣ In this relationship both species benefits.
- ▣ Ex. Ants and aphids
- ▣ Aphids are small insects that suck fluids from the phloem of living plants with their piercing mouth parts.
- ▣ They excrete them through their anus.
- ▣ Ants carry the aphids to new plants and then consume the excreted materials of aphids as food.
- ▣ So, Aphids get new plants and ants get food



Parasitism

- ▣ The parasite is much smaller than the prey that closely associated with it.
- ▣ This relationship is harmful to the prey organism and beneficial to the parasite.

Ex, Lice

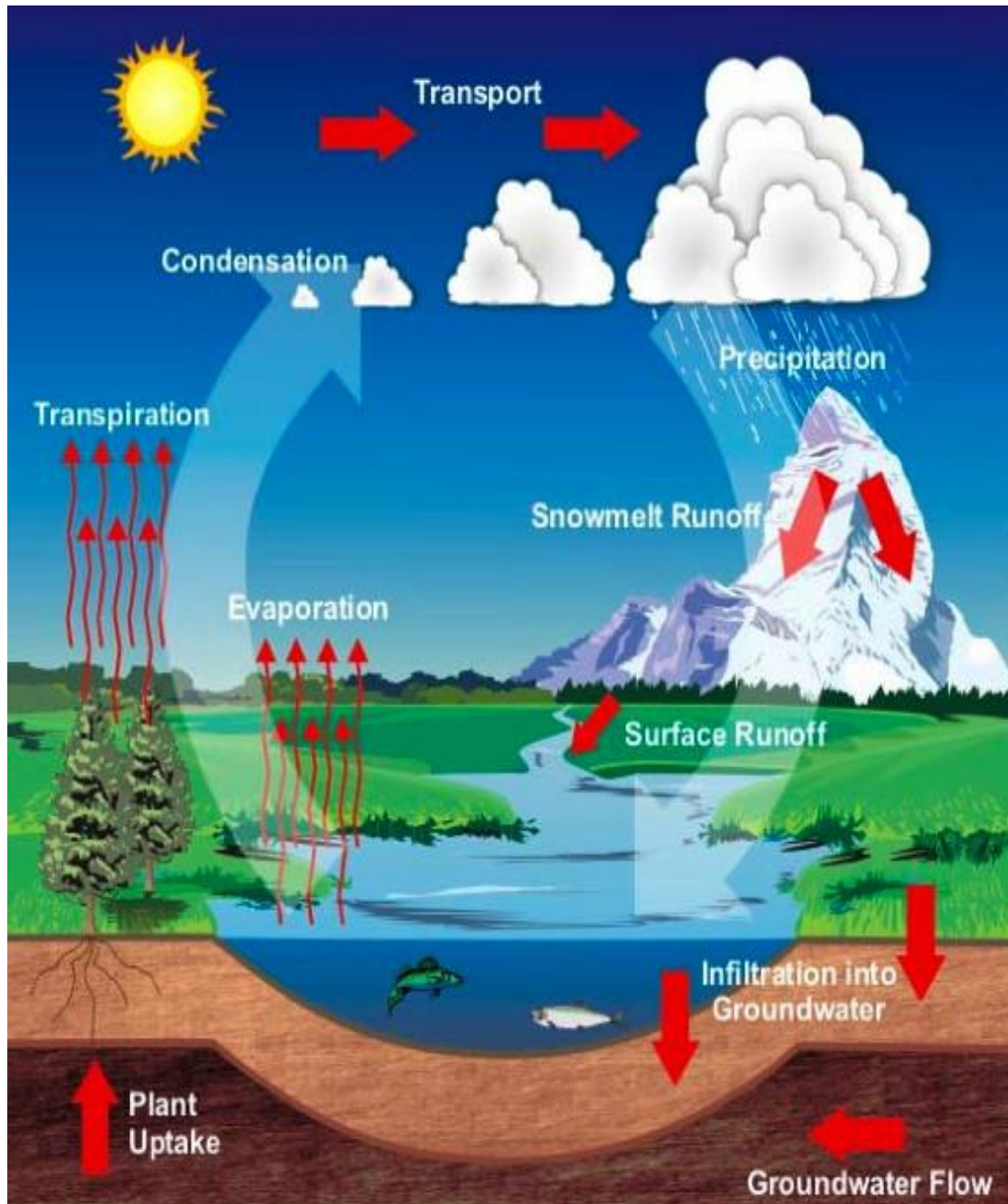
Mites

Ecology 02

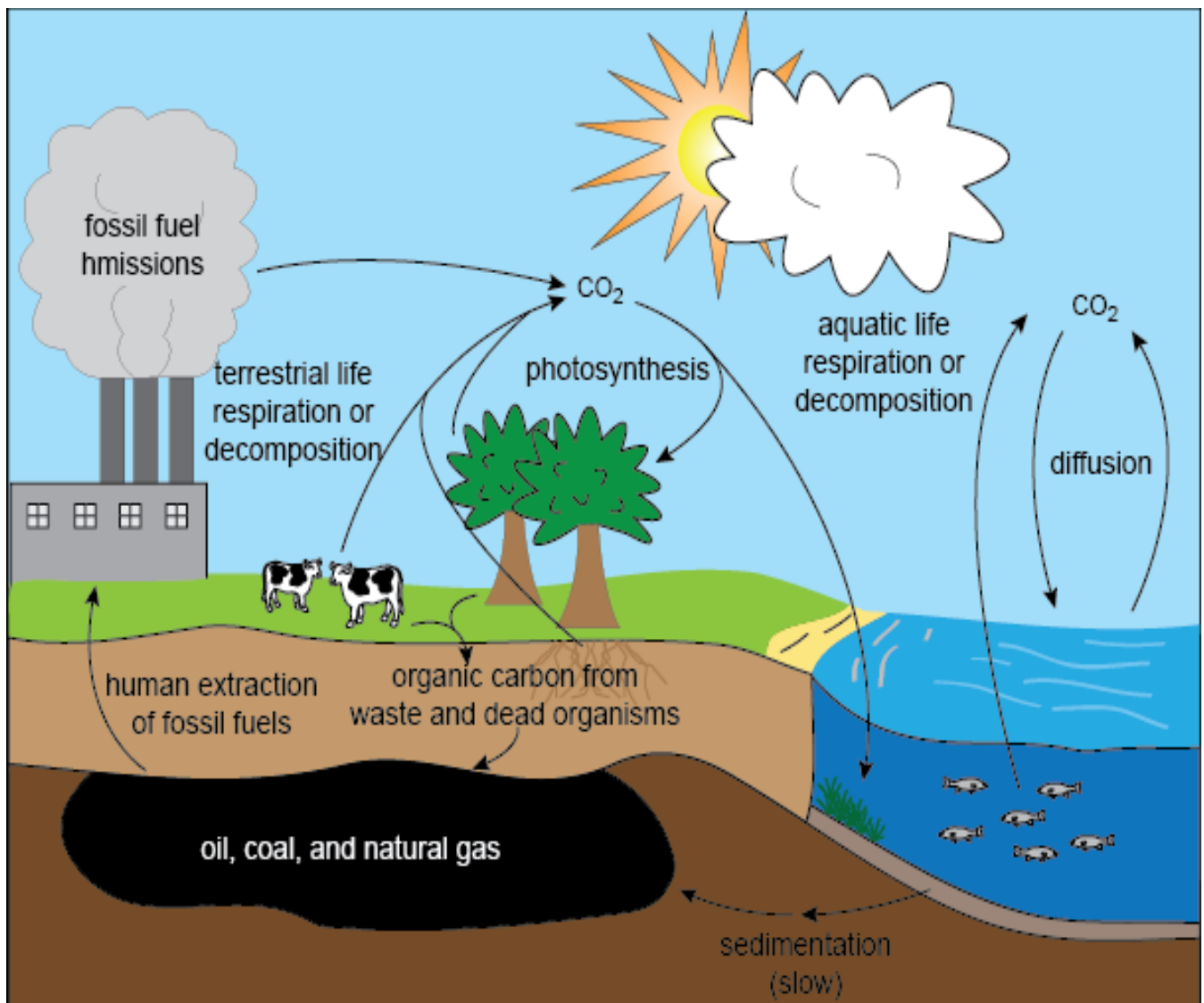
DYNAMICS OF THE Ecosystem

- ▣ Every ecosystem has regulated transfer of energy and a controlled cycling of nutrients.
- ▣ All of the substances cycle through ecosystems in biogeochemical cycles.

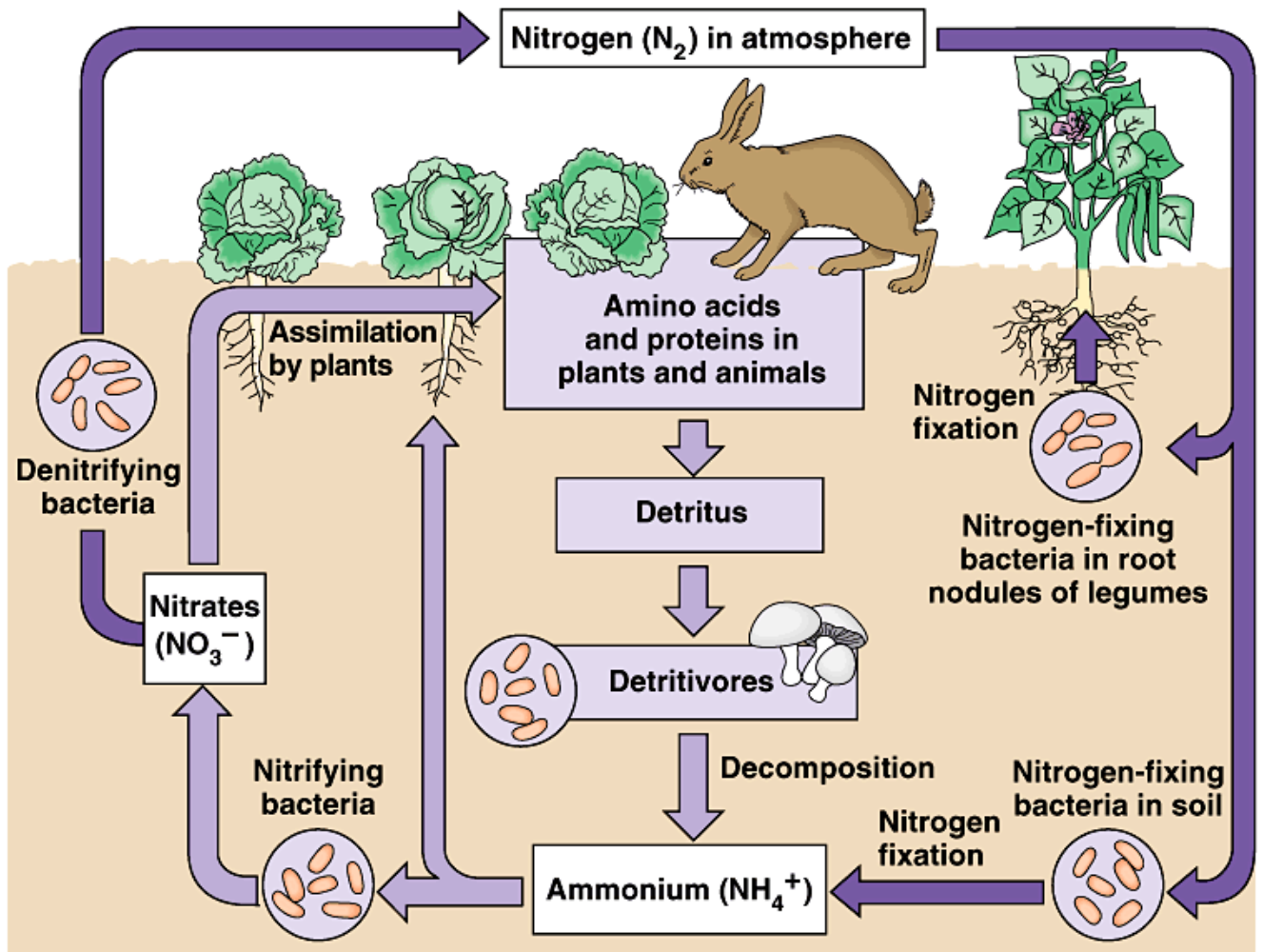
Water cycle



Carbon cycle



Nitrogen cycle

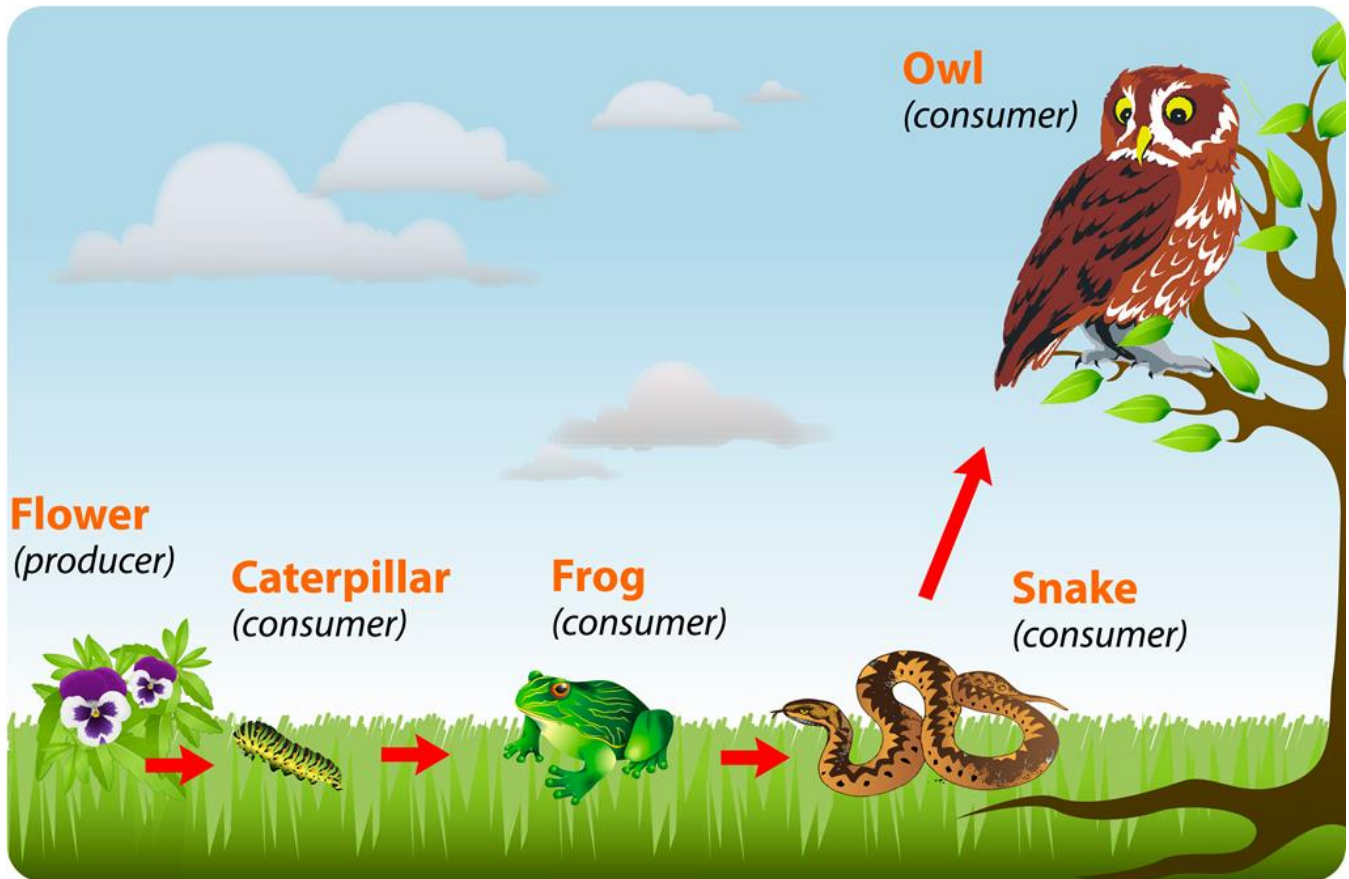


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Trophic levels

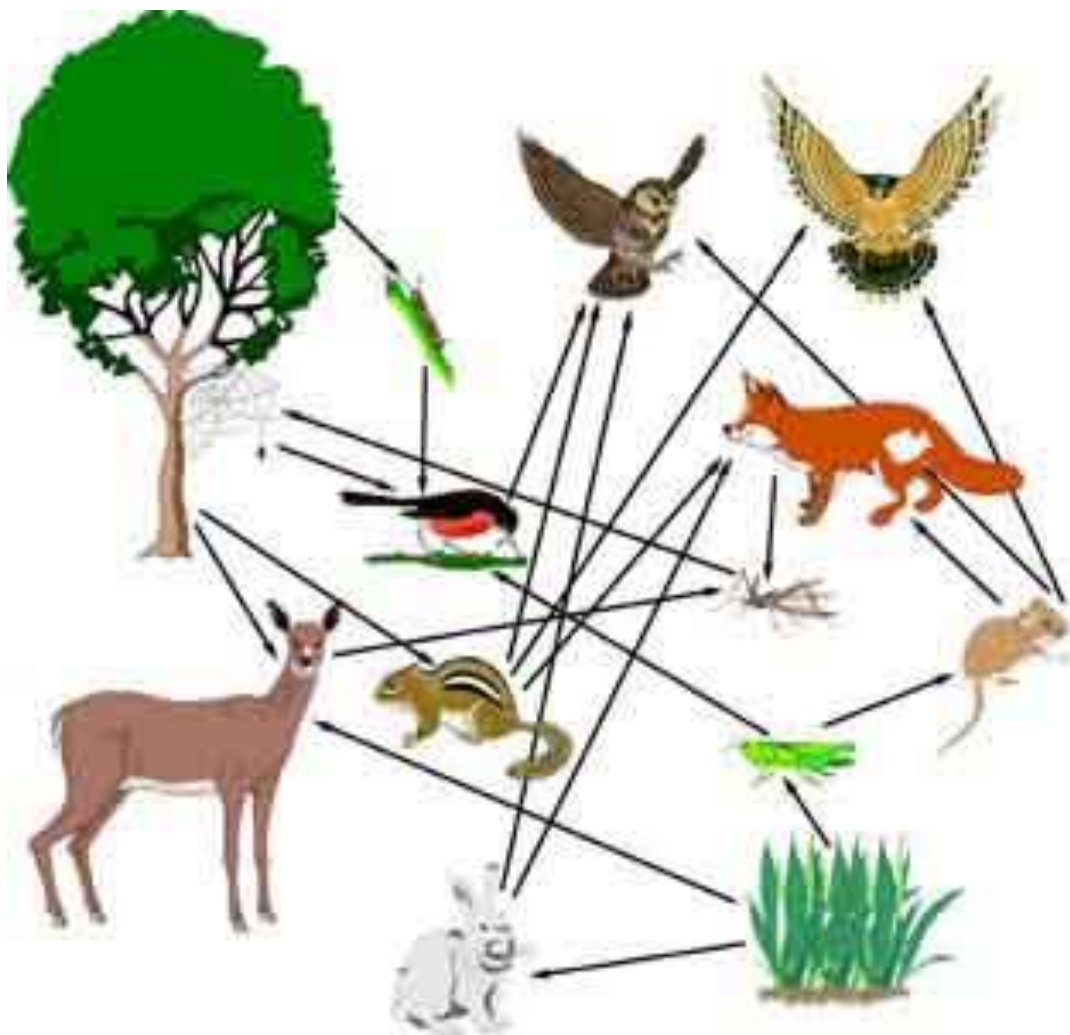
- ▣ An ecosystem is composed of autotrophs and heterotrophs.
- ▣ Autotrophs are the plants, algae and some bacteria that are able to capture light energy and manufacture their own food.
- ▣ They are primary producers.
- ▣ Heterotrophs are the animals, fungi and most other organisms, who obtain organic molecules that have been synthesized by autotrophs.
- ▣ They are consumers.
- ▣ There are several different levels of consumers.
- ▣ Primary consumers are herbivores.
- ▣ They feed directly on the green plants.
- ▣ Secondary consumers are carnivores and parasites of the animals feed in and on the herbivores.
- ▣ Decomposers breakdown the organic matter accumulated in the bodies of other organisms.
- ▣ All of these categories are occurred in any ecosystem.
- ▣ They represent different “Trophic levels”
- ▣ Organisms from each trophic level feed on one another make up a series called “food chain”

An example for a food chain,



- ▣ In real life it is rare for a given kind of organism to feed only on one other type of organism.
- ▣ Usually each organism feed on 2 or more kinds and it is eaten by several other kinds of organisms.
- ▣ When diagrammed it appears as a series of branching lines, than a straight line.
- ▣ It is called a food web.

An example for a food web,



Ecology 03

BIOMES

- Biome is the largest terrestrial community or the largest ecological unit.
- It has specific kinds of plants and animals
- It has distinct climatic conditions.
- Biomes are distributed on earth as broad belts

The major biomes

1. Tundra

- ❖ Found across Asia, Europe and North America.
- ❖ Rainfall below 25cm
- ❖ Poor light and extremely low temperature.
- ❖ Snowfall is scanty
- ❖ Sparse vegetation.
- ❖ Dwarf willow trees, lichens, mosses are found.



2. Taiga

- ❖ It is occurred at the east west band across North America, Europe and Asia.
- ❖ Rainfall 10-35cm
- ❖ Average temperature 6°C in winter and 20 °C in summer
- ❖ Evergreen trees



3. Deciduous forest

- ❖ Occurred in temperate region of North Central Europe, Easton Asia and Easton United states.
- ❖ Rain fall 75-150cm
- ❖ Broadleaved hardwood trees are found



4. Tropical rain forest

- ❖ Occurred in equatorial regions
- ❖ Rainfall exceeds 140 cm
- ❖ Warm and humid climate
- ❖ Luxuriant and varied vegetation



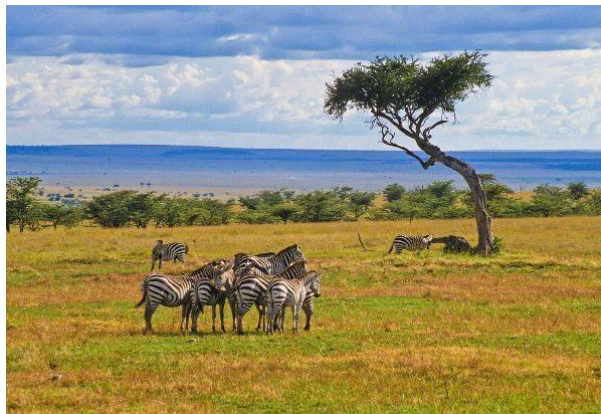
5. Chaparral

- ❖ Extends along Mediterranean Sea, Pacific coast of South America, Chili, South Africa and South Australia.
- ❖ Limited winter rain followed by drought.
- ❖ Vegetation composed of broad-leaved evergreen shrubs, bushes, and small trees usually less than 2.5 m tall; together they often form dense thickets



6. Tropical savannah

- ❖ South America, Africa, Asia and Australia
- ❖ Warm climate
- ❖ Rain fall 100-150 cm



7. Grasslands

- ❖ Prairies of Canada and USA, Pampas of South America, Steppes of Europe and Asia, Veldts of Africa
- ❖ Rainfall is about 25-75cm
- ❖ Short and tall grasses found



8. Desserts

- ❖ South western USA, Mexico, coastal area of Chili and Peru, North Africa
Sahara dessert
- ❖ Extremely low rainfall. Less than 25cm, high evaporation.
- ❖ Hardy grasses found.



Ecology 04

Pollution and Control

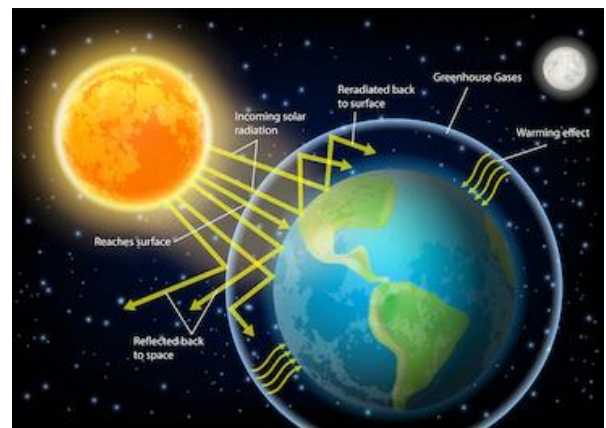
- Pollution is an undesirable and excessive addition of substances to water, air and land that adversely alter the natural quality of the environment.
- The agent that pollute is called pollutant.

Air pollution

- The occurrence of many foreign materials or gases in the atmosphere may be harmful to man, animals and vegetation
- Mainly they are in 2 categories

1. Gaseous

- ▣ Remain in gaseous state in normal temperature and pressure.



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Examples

1. CO₂

- Not harmful gas by itself.
- It is released to the atm. by burning of fossil fuel.
- It forms a blanket around the earth.
- It stops the heat from escaping and causes temperature to rise called “greenhouse effect”.

2. Oxides of Sulphur

- Both SO_2 and SO_3 are harmful.
- They react with water and form Sulphur acid which cause acid rains when precipitate.
- It is harmful to plants, fishes etc.

3. Oxides of Nitrogen

- They form dilute solution of nitric acid cause acid rains.
- It is harmful for human.

4. CO

- Colorless, odorless, nonirritant, but highly toxic gas.
- Diffuse to the blood stream and combine with Hemoglobin and stops O_2 transport inside the body.

2. Particulate

- They are the substances that occur in the solid or liquid form

Example,

Smog

- Incomplete combustion of petrol causes smoke.
- It is mainly a suspension of carbon particles.
- It reduces the amount of sunlight reaching the earth.
- When it combines with water vapor, it causes “smog”.
- It causes lung disorders in human.

Control of Air pollution

▣ Before releasing the gasses into atmosphere from various industries,

1. Separate the pollutants from harmless gasses.
2. Convert the pollutants in unarmful products.

Water pollution

1. Compounds of Hg, As, Pb

- These are released by the industries as iron and steel industry.
- Cause skin disease, and irreversible brain damage.

2. Inorganic Nitrates and Phosphates

- These are released by the chemical plants and cause eutrophication
- It is the condition of sudden increase of nutrients in lakes.
- It affect fauna and flora.



3. Organochlorins

- These are released by the excess use of chemicals for agriculture.
- They pass through food chains.
- And Affect nervous system

4. Animal excreta

- They are released with untreated sewage.
- It contains pathogens and toxic wastes.
- It causes epidemic diseases like cholera.

5. Oils in sea water

- Are released by oil tankers when being loaded or unloaded.
- It kills sea birds. Plants and animals in oceans.

6. Hot water

- Is released by electric power plants
- In lakes it causes sudden increase in temperature in the water body, affect fauna and flora.

Control of water pollution

- ▣ 1. Sewage and factory wastes should be treated before flow into water ways.
- ▣ 2. Various pollutants should be recycled instead of dumping in to water.

Last Lecture

NUTRITION

Autotrophic and Heterotrophic Nutrition

- ✓ Living organisms obtain chemical energy, they need to perform activities, in one of two ways.

Autotrophic

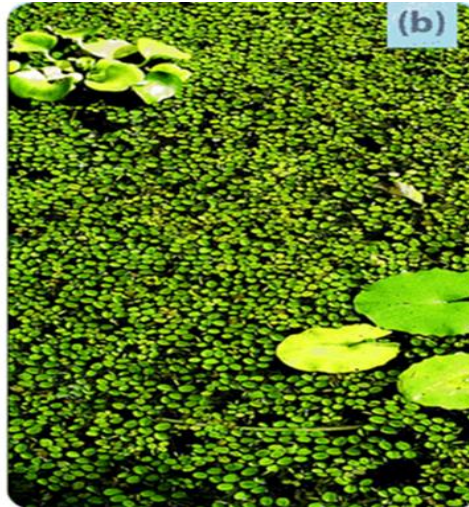
- ✓ Autotrophs store chemical energy in carbohydrate food molecules they build themselves.
- ✓ Food is chemical energy stored in organic molecules.
- ✓ Food provides both the energy to do work and the carbon to build bodies.
- ✓ Because most autotrophs transform sunlight to make food, we call the process they use **photosynthesis**.
- ✓ Only three groups of organisms - plants, algae, and some bacteria - are capable of this life-giving energy transformation.
- ✓ Autotrophs make food for their own use, but they make enough to support other life as well.
- ✓ Almost all other organisms depend on these three groups for the food they produce.

- ✓ The **producers**, as autotrophs are also known, begin **food chains** which feed all life.

a) Plants



b).Algae



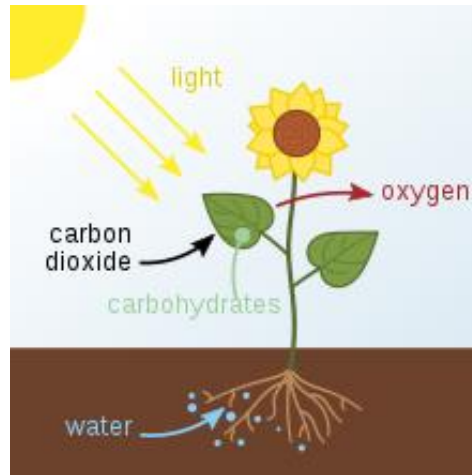
c).Photosynthetic bacteria



- ✓ **Heterotrophs** cannot make their own food, so they must eat or absorb it.
- ✓ For this reason, heterotrophs are also known as **consumers**.
- ✓ Consumers include all animals and fungi and many protists and bacteria.
- ✓ They may consume autotrophs or other heterotrophs or organic molecules from other organisms.
- ✓ Heterotrophs show great diversity but heterotrophs are limited by our utter dependence on those autotrophs that originally made our food.
- ✓ If plants, algae, and autotrophic bacteria vanished from earth, animals, fungi, and other heterotrophs would soon disappear as well.
- ✓ All life requires a constant input of energy.
- ✓ Only autotrophs can transform that ultimate, solar source into the chemical energy in food that powers life.

Photosynthesis

Photosynthesis is the process used by plants, algae and certain bacteria to capture energy from sunlight and turn it into chemical energy.



The process of Photosynthesis

There are two types of photosynthetic processes:

- i. oxygenic photosynthesis
- ii. An oxygenic photosynthesis.

The general principles are very similar, but oxygenic photosynthesis is the most common.

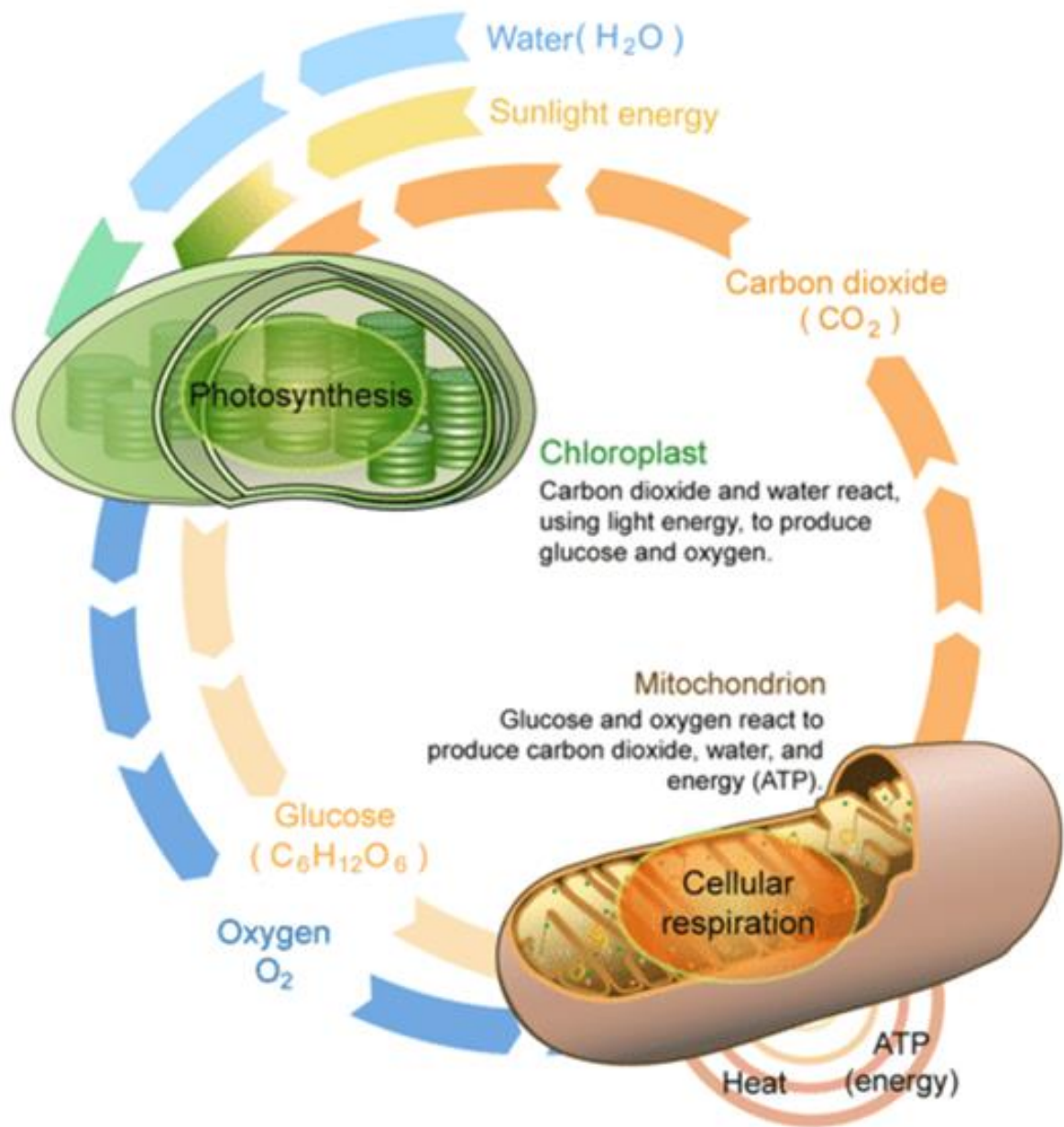
During oxygenic photosynthesis,

- ✓ Light energy transfers electrons from water (H_2O) to carbon dioxide (CO_2), to produce carbohydrates.
- ✓ In this transfer, the CO_2 is "reduced," or receives electrons,
- ✓ And the water becomes "oxidized," or loses electrons.
- ✓ Ultimately, oxygen is produced along with carbohydrates. The capturing of sunlight is done by chloroplast.

- ✓ Anoxygenic photosynthesis does not produce oxygen hence the name is given as anoxygenic.
- ✓ What is produced depends on the electron donor.
- ✓ For example, many bacteria use the bad-eggs-smelling gas hydrogen sulfide, producing solid sulfur as a byproduct.
- ✓ Though both are complex, multistep affairs, the overall process can be neatly summarized as a chemical equation.
- ✓ Oxygenic photosynthesis is written as follows:
 - $6\text{CO}_2 + 12\text{H}_2\text{O} + \text{Light Energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$
- ✓ Here, six molecules of carbon dioxide (CO_2) combine with 12 molecules of water (H_2O) using light energy.
- ✓ The end result is the formation of a single carbohydrate molecule ($\text{C}_6\text{H}_{12}\text{O}_6$, or glucose) along with six molecules each of breathable oxygen and water.
- ✓ Similarly, the various anoxygenic photosynthesis reactions can be represented as a single generalized formula:
 - $\text{CO}_2 + 2\text{H}_2\text{A} + \text{Light Energy} \rightarrow [\text{CH}_2\text{O}] + 2\text{A} + \text{H}_2\text{O}$
- ✓ The letter A in the equation is a variable and H_2A represents the potential electron donor. For example, A may represent sulfur in the electron donor hydrogen sulfide (H_2S).

Using food – Respiration

- ✓ The flow of energy through living organisms begins with photosynthesis.
- ✓ This process stores energy from sunlight in the chemical bonds of glucose.
- ✓ By breaking the chemical bonds in glucose, cells release the stored energy and
- ✓ Make the ATP they need.
- ✓ The process in which glucose is broken down and ATP is made is called cellular respiration
- ✓ Photosynthesis and cellular respiration are like two sides of the same coin.
- ✓ The products of one process are the reactants of the other.
- ✓ Together, the two processes store and release energy in living organisms.
- ✓ The two processes also work together to recycle oxygen in Earth's atmosphere.



Cellular Respiration

- ✓ Cellular respiration actually “burns” glucose for energy.
- ✓ This is because it releases the energy in glucose slowly, in many small steps.
- ✓ It uses the energy that is released to form molecules of ATP.
- ✓ Cellular respiration involves many chemical reactions, which can be summed up with this chemical equation:
- ✓ $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Chemical Energy (in ATP)}$

- ✓ Cellular respiration occurs in the cells of all living things. It takes place in the cells of both autotrophs and heterotrophs.

Reproduction

- ✓ Organisms reproduce to pass on their genes and create new members of their species.
- ✓ If the organisms of a species all fail to reproduce then the species may become extinct.
- ✓ 2 modes of reproduction
- ✓ Asexual and Sexual.

Asexual Reproduction

- ▣ Does not need the union of sex cells or gametes.
- ▣ Offspring produced is usually a clone of the parent.

The different types,

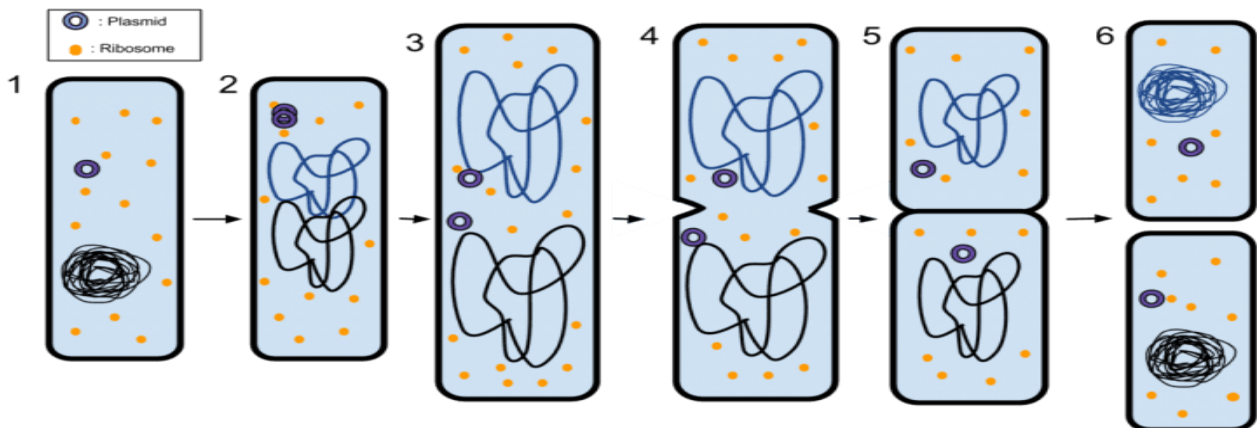
- ▣ *binary fission*,
- ▣ *budding*,
- ▣ *vegetative propagation*,
- ▣ *spore formation (sporogenesis)*,
- ▣ *fragmentation and*
- ▣ *Parthenogenesis*.

Ex;

bacteria, archaea, many plants, fungi, and certain animals.

1. Binary Fission

- A cell divides to produce two identical cells.
- Each of these two cells has the potential to grow to the size of the original cell.
- Ex. Prokaryotes, certain protozoans

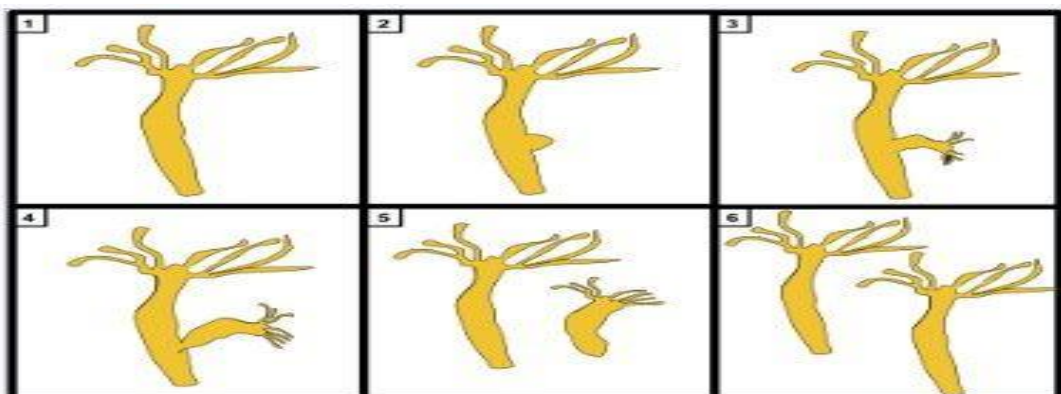


2. Budding

- The formation of an outgrowth (or bud) from an organism
- That is capable of developing into a new individual.
- The outgrowth is genetically the same as the parent but relatively smaller.
- It may stay attached or eventually split off from the parent.

Ex;

Certain bacteria, fungi, and certain asexual animals, such as hydra and corals.

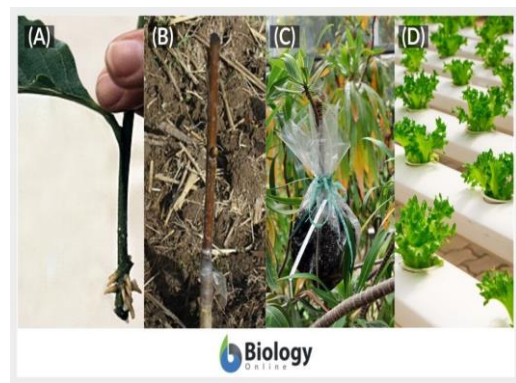


3. Vegetative Propagation

- It is when a new plant emerges from vegetative parts, such as specialized stems, leaves, and roots.
- Then, they form their own root system and grow.

Ex;

those emerging from runners, bulbs, tubers, corms, suckers (root sprouts), and plantlets.



4. Spore Formation (Sporogenesis)

- Involves spores.
- Spores are dormant, reproductive cells and are thick-walled and highly resistant to various unfavorable conditions, like high temperatures and low humidity.
- When the conditions are suitable, they germinate to give rise to new individuals.

Ex; Ferns

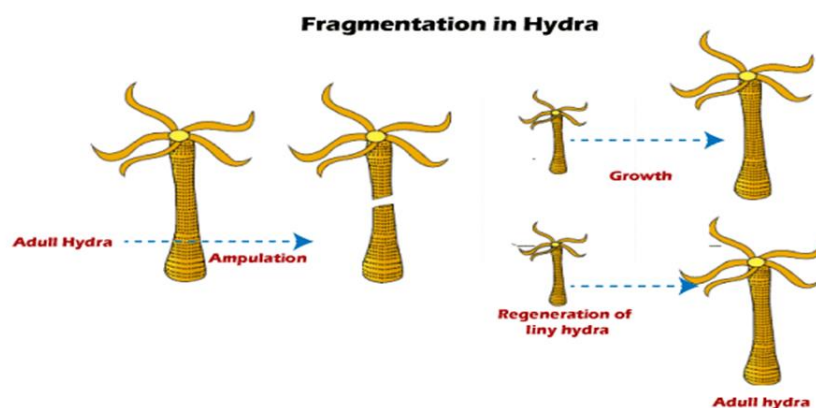


5. Fragmentation

- parent organism breaking into fragments and
- each fragment is capable of developing into a new organism
- Human activity, predation, and other environmental factors may cause them to split into fragments.

Ex;

fungi, molds, vascular and nonvascular plants, cyanobacteria, and animals (e.g. sponges, sea stars, planarians, and many annelid worms).



6. Parthenogenesis

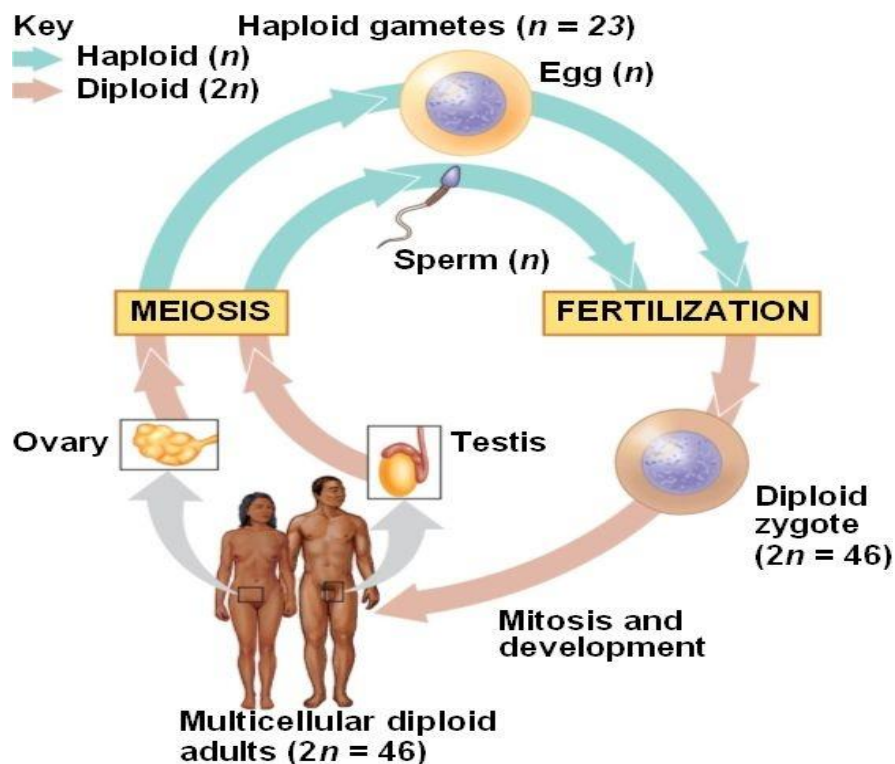
- Offspring develops from a female gamete even without prior fertilization by a male gamete.
- the females lay unfertilised eggs which hatch into females which also lay unfertilised eggs and so on

Ex: insects (ants, bees)



Sexual Reproduction

- Fusion of male and female gametes is a must
- The semen containing sperm cells is released by the male into the female's reproductive organ
- Only a single sperm would be able to fertilize a viable ovum
- The haploid (n) sex cells form the diploid zygote that will next undergo mitosis to become an embryo
- The embryo then develops organs and become a fetus inside the female womb.
- The fetus receives nutrients inside the mother's womb via an umbilical cord.
- After the gestation period (usually, about 266 days), the female gives birth by pushing the fetus out of the birthing canal.
- The newly born child, then, receives nutrition by lactation.
- Postnatal care continues until the child becomes independent.



- This is in contrast to asexual reproduction where an organism reproduces without involving gametes and the resulting offspring is a clone of the parent.
- Nonetheless, sexual reproduction has the advantage over asexual reproduction in increasing genetic variation and expanding the gene pool.
- Furthermore, it ensures that the chromosome number of a particular species will remain the same across generations.

END

Create By Buddika Sampath